

Two Forms of a Linear Equation

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

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|---|---|
| 1. slope-intercept form = $18 - \frac{3x}{2}$ | 7. (a) x-intercept = 3, (b) y-intercept = 6, (c) slope = -2 |
| 2. slope-intercept form = $8 - \frac{4x}{5}$ | 8. (a) drop in large drinks = -9 , (b) slope-intercept form = $12 - \frac{3x}{4}$ |
| 3. constant in standard form = 24 | 9. constant in standard form = 14 |
| 4. (a) x-intercept = 12, (b) y-intercept = 15 | 10. (a) slope-intercept form = $20 - \frac{4x}{3}$, — |
| 5. slope-intercept form = $2x - 7$ | |
| 6. constant in standard form = 20 | |
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What is verified

9 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.EE.B.6 – problems 1, 2, 4, 5, 7, 10

8.F.B – problems 3, 6, 8, 9

Difficulty 1–4 of 5 across 15 tagged checks.

1. $3x + 2y = 36$ (notebooks at \$3, pens at \$2, \$36 total). Rewrite solved for y .

Solution: Subtract the x -term from both sides: $2y = 36 - 3x$. Now divide *every* term by

- 2: $y = 18 - \frac{3}{2}x$. Writing the x -term first gives the usual $y = mx + b$ order. $y = -\frac{3}{2}x + 18$

2. $4x + 5y = 40$ (child tickets \$4, adult tickets \$5, \$40 total). Rewrite solved for y .

Solution: $5y = 40 - 4x$, then divide every term by 5: $y = 8 - \frac{4}{5}x$.

$y = -\frac{4}{5}x + 8$

3. $y = -\frac{2}{3}x + 8$. Rewrite as $2x + 3y = C$.

Solution: Substitute the expression for y into $2x + 3y$:

$$2x + 3\left(-\frac{2}{3}x + 8\right) = 2x - 2x + 24 = 24.$$

The x -terms cancel, which is exactly why the standard form has a constant on the right.

$2x + 3y = 24$

4. $5x + 4y = 60$ (posters \$5, flyers \$4, budget \$60). (a) x -intercept. (b) y -intercept.

Solution (a): At the x -intercept, $y = 0$, so $5x = 60$.

$$x = 12$$

Solution (b): At the y -intercept, $x = 0$, so $4y = 60$.

$$y = 15$$

5. $2x - y = 7$. Rewrite solved for y .

Solution: Subtract $2x$: $-y = 7 - 2x$. Multiply both sides by -1 — every term changes sign: $y = 2x - 7$.

$$y = 2x - 7$$

6. $y = \frac{3}{4}x - 5$. Rewrite as $3x - 4y = C$.

Solution: Substitute into $3x - 4y$:

$$3x - 4\left(\frac{3}{4}x - 5\right) = 3x - 3x + 20 = 20.$$

$$3x - 4y = 20$$

7. $6x + 3y = 18$ (seeds \$6, soil \$3, fund \$18). (a) x -intercept. (b) y -intercept. (c) Slope from the two intercept points.

Solution (a): Set $y = 0$: $6x = 18$.

$$x = 3$$

Solution (b): Set $x = 0$: $3y = 18$.

$$y = 6$$

Solution (c): The two intercept points are $(3, 0)$ and $(0, 6)$. Going from the first to the second, y rises by 6 while x falls by 3, so the slope is $6/(-3)$.

$$m = -2$$

8. $3x + 4y = 48$ with the table $x = 0, 4, 8, 12$ against $y = 12, 9, 6, 3$. (a) Change in y from $x = 0$ to $x = 12$. (b) Slope-intercept form.

Solution (a): The last table entry minus the first: $3 - 12$.

$$-9$$

Solution (b): $4y = 48 - 3x$, then divide every term by 4: $y = 12 - \frac{3}{4}x$. Notice the table agrees: 12 small drinks cost 9 large ones, which is a drop of $\frac{3}{4}$ per small drink.

$$y = -\frac{3}{4}x + 12$$

9. $y = -\frac{5}{2}x + 7$. Rewrite as $5x + 2y = C$.

Solution: Substitute into $5x + 2y$:

$$5x + 2\left(-\frac{5}{2}x + 7\right) = 5x - 5x + 14 = 14.$$

$$5x + 2y = 14$$

10. $8x + 6y = 120$ (tomatoes \$8, peppers \$6, budget \$120). (a) Slope-intercept form. (b) What the slope means for the gardener.

Solution (a): $6y = 120 - 8x$, then divide every term by 6: $y = 20 - \frac{8}{6}x$, and $\frac{8}{6}$ reduces to $\frac{4}{3}$.

$$y = -\frac{4}{3}x + 20$$

Solution (b) — instructor-judged. A correct explanation reads the slope as a trade-off in the garden: each extra tomato plant costs four thirds of a pepper plant, or equivalently every 3 extra tomato plants cost 4 pepper plants, and the negative sign says one number falls as the other rises. The units may be stated either way round. “The slope is negative” with no reference to plants does not earn credit.